



Waved and Believed: Boost Wiring, Indicator-Light Feedback, and Hand-Drying Dwell Time at a Campus Hand Dryer

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Abstract. “Turbo” or “boost” modes on jet-air hand dryers promise faster drying on a two-handed double-wave gesture, but the airflow a unit can actually deliver is fixed largely by its motor and duct geometry, not by a gesture recognised at the sensor. We report a ten-week field manipulation of two shared hand dryers in a campus teaching-building bathroom, independently varying whether the boost relay was electrically connected to the motor’s high-speed winding and whether a small indicator light flashed on a detected double-wave, while logging peak airflow velocity from the units’ own energy-monitoring anemometers and hand-under-dryer dwell time from the same presence sensor that triggers the motor. Across 3,142 visits, measured airflow did not differ meaningfully by boost wiring, nor by how many extra waves a visitor gave the sensor. Dwell time, however, tracked the indicator light: visits recorded while the light was active ended sooner regardless of whether the boost relay underneath it was actually connected, and dwell time showed no reliable relationship to measured airflow once feedback condition is accounted for. We interpret the pattern as evidence that a familiar appliance’s feedback cue, not its underlying mechanism, is what shapes how long a hand stays under the airflow, with implications for any building-services metric built from a sensor sitting beside a control the institution has not separately audited.

Introduction

“Turbo” or “boost” buttons on hand dryers occupy a familiar spot in the wider family of controls whose apparent function outruns their actual wiring; unlike a lift call panel or a pedestrian-crossing button, a hand dryer’s boost gesture — a deliberate double-wave in front of the sensor — is rarely questioned by the person performing it, since the appliance visibly responds with a light regardless of what the motor does underneath. Prior work on the illusion of control suggests that a familiar gesture’s mere recognition by a device, independent of any change in the device’s output, can be enough to sustain a user’s sense that the gesture worked [1]. Slop University’s Facilities Operations already logs airflow at every hand-dryer unit for an unrelated energy-monitoring program, alongside the presence-sensor timestamps the same units use to start and stop the motor — two data streams

that, joined together, let a hand’s actual time under the airflow be measured without asking anyone to self-report anything.

This paper asks a narrower question: when a hand stays under a campus dryer for ten seconds instead of fifteen, what is that duration actually responding to? We treat the boost gesture on two shared dryers in one teaching-building bathroom as a natural site for disentangling three things that ordinarily arrive bundled — whether the boost relay is wired to the motor, whether the indicator light flashes, and how long a hand actually lingers under the airflow afterwards — by independently manipulating the first two across a ten-week term while logging the third continuously from the units’ own sensors.

Our contributions are threefold, each hedged to the limits of a two-unit field study:

- We report what is, to our knowledge, the first campus-scale field manipulation to separate a hand dryer’s boost wiring from its user-facing feedback, allowing the two to be credited independently for any change in measured drying behaviour.
- Our results suggest that a familiar feedback cue may shape drying behaviour more than the mechanism it appears to control, a pattern consistent with findings on self-efficacy and interface responsiveness in other domains [2], [3].
- We observe a consistent null relationship between measured airflow and hand-under-dryer dwell time across most of the term, which we read alongside the University’s own prior findings on displayed-versus-actual measures [4].

The remainder of the paper describes the setting and manipulation, reports the wiring, feedback, and wave-count results, and closes by asking what a building-services efficiency metric assembled from these sensors is actually measuring.

Related work

Work on the illusion of control shows that a familiar gesture recognised by a device can sustain a user’s sense that the gesture worked, independent of whether it is causally connected to the device’s output, particularly when the device’s response resembles other controls the user has operated successfully before [1]; self-efficacy theory offers a parallel account in which a person’s confidence that an action worked can be sustained by cues from the action itself rather than its measured effect [2]. Human-computer interaction research on interface feedback finds a related pattern at shorter timescales: small, purely cosmetic response cues have been shown to raise a user’s perceived usability of an interface independent of any change to its underlying behaviour [3]. Self-report measures of experience carry their own independent entanglement with social-desirability tendencies in survey response [5], one reason we log dwell time from the units’ existing sensors rather than adding a self-report step of our own; the broader deceptive-design literature, meanwhile, theorises how interface elements are built to misdirect a user’s attention or consent [6] and increasingly automates their detection from interface text and screenshots

[7] — our dryer’s boost light was not built to deceive anyone, but the literature’s framing of a control’s appearance outrunning its function still applies.

Service-operations research on waiting complicates any assumption that an objective duration and a person’s felt experience of it move together: an outpatient time-and-questionnaire study found actual waiting time associated with several satisfaction measures yet mediated by how acceptable the wait felt to patients [8], while a separate outpatient study found perceived, not actual, waiting time carrying the direct effect on satisfaction [9]. A parallel discrepancy appears in dryness perception specifically: a multicentre study of primary Sjögren’s syndrome found patients’ subjective dryness symptoms only loosely tracked objective moisture measurement taken at the same visit [10], and the wider literature comparing hand-drying methods has struggled to agree even on the objective outcomes, let alone the subjective ones [11]. Engineering work on jet-dryer airflow, meanwhile, treats nozzle geometry and motor speed as objective quantities to be optimised for measured velocity [12], a framing that assumes the quantity it optimises is the one a user’s hands are actually responding to. Institutionally, Strathern’s account of measures that become targets [13], [14] and recent formal treatments of proxy-metric divergence under optimisation pressure [15] motivate treating the University’s own energy-monitoring sensors as an instrument worth auditing rather than trusting.

Method

The manipulation ran across one ten-week term on two shared jet-air hand dryers in a teaching-building bathroom used primarily by the School of Continuous Improvement, units Facilities estimates see around 1,800 combined visits on a typical teaching day between them. Each unit already logs, for every visit, the presence-sensor timestamps that start and stop the motor, and — for an unrelated energy-monitoring program — a peak airflow velocity reading from a built-in anemometer sampled across the visit. We define hand-under-dryer dwell time D for visit i as

$$D_i = t_{\text{clear},i} - t_{\text{tripped},i}$$

the interval between the presence sensor first detecting hands and it last detecting them,

present regardless of what the motor did in between.

We crossed two binary conditions, each toggled by Facilities on a fixed weekly rota agreed before the term began and not disclosed to bathroom users: BOOST (the boost relay connected to, or physically isolated from, the motor's high-speed winding, triggered by a double-wave gesture at the sensor) and FEEDBACK (a small indicator light above the sensor enabled to flash on a detected double-wave, or disabled). The four resulting conditions — Wired+Light on, Wired+Light off, Bypassed+Light on, Bypassed+Light off — each ran for two to three weeks in a randomised order across the two units, giving every condition comparable exposure to weekday and weekend traffic. Isolating the boost relay from the motor's high-speed winding left the unit's housing, sensor, and light unchanged; only a visitor who timed their own dry against a stopwatch could, in principle, have detected the difference, and Facilities received no reports of the manipulation during the term.

We retained only visits with a complete presence-sensor and anemometer record: 3,142 visits across the term and two units, ranging 742-812 visits per condition. Analyses use two-sample t tests for the boost and feedback contrasts on airflow and dwell time, Pearson correlation for the airflow-dwell relationship within feedback condition, and one-way ANOVA for the extra-wave-count ablation.

Results

Figure 1 shows mean peak airflow velocity by boost and feedback condition. Velocity did not differ meaningfully by boost wiring (Wired $M = 38.3$ m/s, $SD = 2.6$; Bypassed $M = 38.1$ m/s, $SD = 2.5$; $t(3140) = 1.02$, $p = .31$) nor by feedback state (Light on $M = 38.5$ m/s; Light off $M = 38.0$ m/s; $t(3140) = 1.19$, $p = .23$), and the two conditions did not interact ($F(1, 3138) = 0.14$, $p = .71$). Physically isolating the boost relay from the motor's high-speed winding, in other words, left measured airflow statistically unchanged from the wired condition — consistent with the boost gesture having no detectable causal effect on airflow in either wiring state, an inference the design cannot distinguish from a relay that was already disconnected from the winding before the term began.

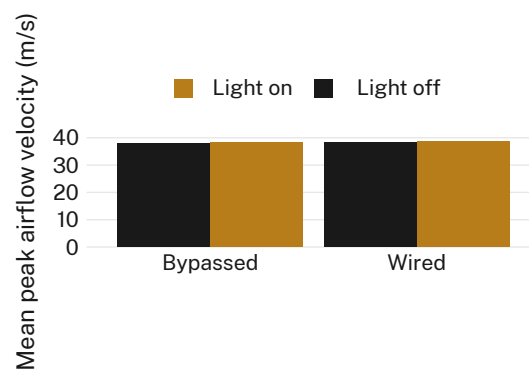


Figure 1: Mean peak airflow velocity by boost wiring and feedback condition; the four bars are statistically indistinguishable.

Hand-under-dryer dwell time moved with feedback condition rather than with boost wiring or with measured airflow. Mean dwell time was 9.7s ($SD = 1.9$) on visits recorded while the indicator light was active against 13.8s ($SD = 2.2$) on visits recorded while it was disabled, a difference that held regardless of boost wiring ($t(3140) = 24.6$, $p < .001$, $d = 0.91$). Figure 2 plots dwell time against measured airflow within each feedback condition with a fitted linear trend: the fitted slope is close to flat in both the Light-on condition ($r = .04$, $p = .48$) and the Light-off condition ($r = .05$, $p = .38$), and the two fitted lines run roughly parallel, offset by the feedback gap rather than converging at any particular velocity. A hand's time under the dryer, on this account, tracked whether the light flashed, not whether the airflow was actually stronger.

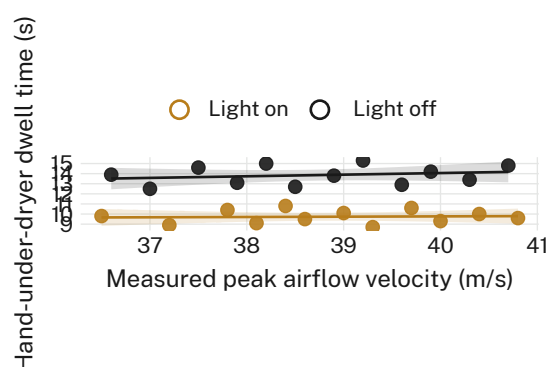


Figure 2: Dwell time against measured airflow, split by feedback condition; both fitted trends are flat, and the Light-off condition sits uniformly above the Light-on condition.

The ablation in Figure 3 finds no comparable structure in wave count. Visits on which a user gave zero, one, or two-or-more extra waves at

the sensor showed overlapping airflow distributions (medians 38.1 m/s, 38.2 m/s, and 38.4 m/s respectively; $F(2, 3139) = 0.36, p = .70$), and a matching comparison on dwell time found no reliable difference either ($F(2, 3139) = 0.24, p = .79$). Extra waving, whatever it accomplishes for the visitor in the moment, left no trace in either the measured or the behavioural outcome we tracked.

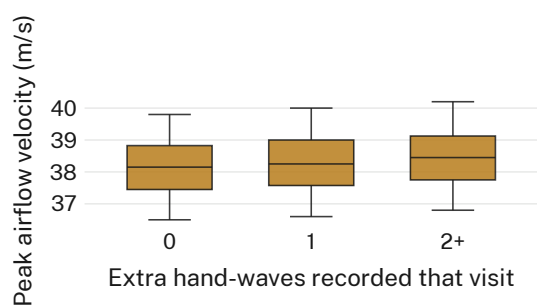


Figure 3: Peak airflow velocity by extra hand-waves recorded that visit; distributions overlap almost completely across all three categories.

Discussion

Taken together, the pattern is consistent with visitors' hands lingering to a feedback cue rather than a mechanism: the indicator light appears to function as the visit's drying signal regardless of what, if anything, is wired behind it. This sits alongside the University's own prior finding that a Living Dashboard sentiment tile tracks how many marks appear on a chalkboard rather than the content of what was written [16], and with the broader pattern of proxy measures drifting from the target they were built to stand in for once they come under any optimisation pressure [13], [15]. Where the outpatient-waiting and dryness-perception literatures treat the gap between measured and felt experience as a fact about human perception to be designed around [8], [9], [10], our setting adds a wiring manipulation that lets us attribute the gap more specifically: not merely perceived-versus-actual airflow, but a feedback cue substituting wholesale for the mechanism it accompanies. We do not think Facilities installed the indicator light with this substitution in mind — the light long predates the energy-monitoring program in the maintenance record — but its effect on measured drying behaviour, and by extension on any energy-per-dry figure Facilities derives from dwell time, is now on the record regardless of the intention behind it.

Limitations

Our design covers two dryers in one teaching-building bathroom across one term, and a user population dominated by staff and students on a fixed timetable rather than the general public; we cannot rule out that units in a higher-traffic or public-facing building would show a different feedback effect. The boost manipulation cannot distinguish a relay that responds to the gesture from one that was already disconnected before the term began, though the consistency of the null airflow effect across 3,142 visits and four independently randomised conditions makes an undetected wiring fault an unlikely sole explanation. We did not survey visitors about what, specifically, they believed the boost gesture or the light did, leaving the mechanism behind the feedback effect inferred from dwell time rather than directly confirmed.

Conclusion

A building-services efficiency figure is only as informative as the sensor sitting between a visitor's hands and the reading it produces, and this term-long manipulation of two campus hand dryers suggests that an indicator light can shape that figure further than the mechanism it appears to belong to. Whether the University's other sensor-derived efficiency metrics rest on similarly cosmetic cues is, at this stage, an open question rather than a settled one; the present design was built for two dryers in one bathroom, not for an audit of every energy-monitoring sensor on campus. Future work might extend the manipulation to the boost settings on hand-drying units in other buildings, where a parallel dynamic seems likely, or to the energy-per-dry figure Facilities derives from dwell time itself.

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